

BACH LONG VI, Viet Nam

WMO: 488390

Lat: 20.130N

Lon: 107.720E

Elev: 184

StdP: 14.60

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	52.3	53.9	40.0	36.7	57.7	43.4	41.8	58.0	38.5	56.6	36.1	56.5	24.5	20	0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.3	91.5	83.7	90.3	83.2	89.3	82.8	84.5	89.8	83.8	88.8	83.3	88.0	12.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
83.1	173.7	88.3	82.5	170.5	87.6	81.9	167.1	87.0	49.1	90.3	48.3	89.3	47.6	88.3	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.3	31.8	27.9	DB	51.2	94.3	2.4	2.8	49.5	96.4	48.1	98.0	46.7	99.6	45.0	101.7	(4)
(5)			WB	46.3	85.8	2.4	0.9	44.6	86.5	43.2	87.0	41.9	87.5	40.2	88.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	75.8	63.8	64.3	67.7	75.0	81.1	84.9	85.1	84.2	82.3	79.2	74.0	67.6		
	DBStd	8.89	5.62	6.42	5.41	5.05	3.39	2.38	2.52	2.41	2.56	3.05	4.59	5.07		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	247	91	85	35	2	0	0	0	0	0	0	3	31		
	CDD50	9429	428	402	548	749	966	1048	1087	1060	970	905	721	545		
	CDD65	4201	55	67	118	301	501	598	622	595	520	440	273	111		
	CDH74	40694	28	88	236	1703	5000	7528	7930	7321	5791	3677	1197	195		
	CDH80	13972	0	1	9	255	1450	3276	3529	2943	1772	595	130	12		
(14) Wind	WSAvg	13.6	15.1	13.9	12.8	12.1	12.8	14.5	14.7	11.2	11.7	13.1	15.0	16.4		
(15) Precipitation	PrecAvg	46.0	1.1	0.7	1.2	1.5	4.1	6.2	6.5	10.2	8.9	3.4	1.7	0.9		
	PrecMax	67.4	11.1	3.5	3.7	4.0	13.3	13.5	21.4	28.6	18.3	8.6	6.8	2.8		
	PrecMin	28.7	0.0	0.0	0.1	0.0	0.5	0.3	1.0	3.9	1.5	0.0	0.0	0.0		
	PrecStd	10.1	2.2	0.9	1.0	1.2	3.1	3.7	4.8	5.7	4.3	2.5	1.6	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.1	79.3	81.2	87.2	90.6	92.9	92.8	92.0	90.6	87.9	85.6	81.5		
		MCWB	71.8	74.2	76.1	80.5	83.3	83.8	84.2	83.8	84.0	80.1	78.5	76.1		
	2%	DB	74.5	76.8	78.9	84.6	88.8	91.1	91.6	90.6	88.9	86.0	83.4	78.4		
		MCWB	70.0	72.8	74.5	79.1	82.7	83.6	83.7	83.3	82.7	79.6	77.5	73.9		
	5%	DB	72.4	74.5	76.7	82.8	87.4	89.9	90.3	89.3	87.5	84.6	81.2	76.1		
		MCWB	69.0	71.3	73.3	78.1	82.0	83.2	83.0	82.8	81.7	78.6	76.0	72.5		
	10%	DB	70.6	72.5	74.7	81.1	86.0	88.6	89.0	88.0	86.3	83.2	79.2	73.9		
		MCWB	67.9	70.2	72.1	77.2	81.2	82.7	82.5	82.2	80.9	77.7	74.8	70.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.6	74.9	76.6	81.0	84.5	85.3	85.4	85.0	84.6	81.8	79.9	77.1		
		MCDB	75.9	78.6	80.1	85.6	88.9	90.1	91.4	90.5	89.4	86.0	84.2	80.4		
	2%	WB	70.9	73.3	75.0	79.7	83.3	84.4	84.3	84.0	83.2	80.4	78.3	75.0		
		MCDB	73.5	76.1	77.8	83.7	87.5	89.4	90.0	89.2	87.7	84.6	82.0	77.3		
	5%	WB	69.7	71.9	73.7	78.7	82.4	83.7	83.5	83.2	82.3	79.4	77.2	73.3		
		MCDB	71.7	74.0	76.1	82.1	86.4	88.4	88.9	88.0	86.4	83.3	80.1	75.9		
	10%	WB	68.4	70.6	72.4	77.7	81.6	83.2	82.8	82.5	81.5	78.6	76.1	71.3		
		MCDB	70.2	72.3	74.6	80.5	85.2	87.5	87.8	87.0	85.5	82.2	78.8	73.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	5.9	6.3	7.2	7.1	6.5	6.3	6.4	6.1	5.6	5.7	5.7		
		MCDBR	8.7	8.9	9.2	9.1	8.2	8.0	7.7	7.8	7.5	7.3	8.2	9.0		
	5% WB	MCWBR	5.5	5.4	5.2	4.7	4.3	3.8	3.8	4.1	4.7	4.6	4.9	5.5		
		MCWBR	8.4	8.4	8.7	8.7	7.8	7.3	7.2	7.2	7.2	6.9	8.0	8.9		
(40) Clear-Sky Solar Irradiance	taub		0.622	0.651	0.790	0.731	0.536	0.495	0.473	0.519	0.574	0.675	0.599	0.636		
		taud	1.735	1.696	1.495	1.636	2.144	2.281	2.340	2.185	1.982	1.706	1.840	1.718		
	Edn at Noon		209	213	191	206	248	256	262	252	236	204	213	199		
		Edh at Noon	69	75	95	83	49	43	41	48	57	73	61	68		
(44) All-Sky Solar Radiation	RadAvg		800	944	1050	1418	1717	1727	1696	1563	1488	1384	1123	864		
	RadStd		180	161	137	151	144	160	200	106	134	91	132	129		

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	+0.90	+0.68	+0.68	N/A	N/A	N/A	N/A	(46)
(47)	Regional (4 neighbors)	+0.41	N/A	N/A	+0.67	+0.45	+0.50	N/A	N/A	+135	+135	(47)

Nomenclature: See separate page